INDIA RISING

Although India continues to advance its engineering prowess at an incredible pace, the country is facing multiple challenges in continuing that growth trajectory.

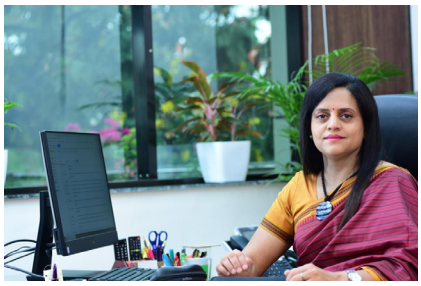
Written by Poornima Apte

ASHWINI BHIDE, the managing director of the Mumbai Metro Rail Corporation, remembers when work on the Aqua Line of the Mumbai Metro project completely stalled.

The ambitious 33.5-km-long (a little more than 20 miles) corridor connects six business districts, 30 educational institutes, 30 recreational facilities, and domestic as well as international airport terminals. It's Mumbai Metro's first underground line and in a densely populated city like Mumbai with around 20 million residents, the construction of tunnels and underground stations was no easy task. One of the tunnels for this project had to be dug underneath a river, further complicating an already challenging engineering project. Practically everything—ensuring non-disruption of utilities, preserving heritage buildings, financing, right-of-way issues, worker safety, media, and communication—was a challenge, but Bhide found workarounds. But the biggest challenge came when, after more than half the project was complete, the government issued a stay on construction. "It put us on a complete back footing because we were neither able to simply drop the

project nor were we able to go ahead because by that time we had already spent a substantial amount of money," Bhide said.

The metro project found a new lease of life with a change in government and Bhide saw the Aqua Line through to completion. This $4 billion project, expected to be completed by October 2025, is an example of India laying the groundwork toward occupying a bigger presence on the global stage. Already a destination for software development, the country is firing on all cylinders, incentivizing "Make in India" manufacturing and developing a healthy ecosystem for startups in the process.



Ashwini Bhide has led Mumbai Metro's first underground line through delays and political changes to near completion. Photo: Mumbai Metro

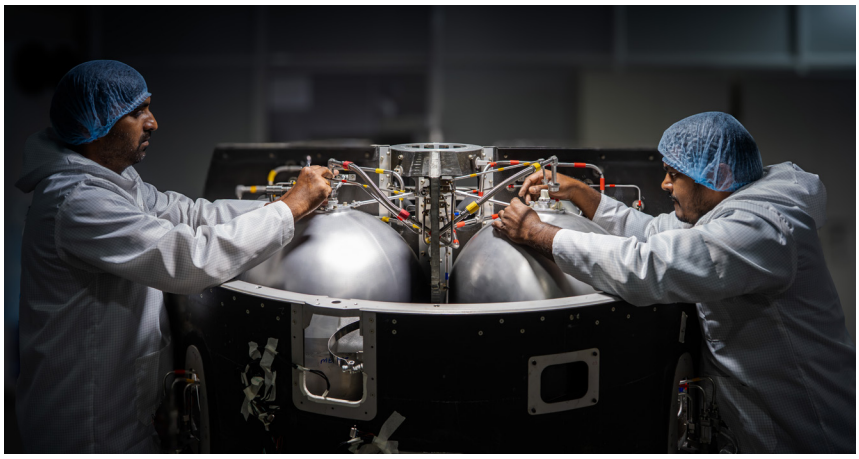
India boasts multiple advantages, including inexpensive labor and a large pool of talent, said Ravi Soni, an expert who worked in the country's manufacturing industry and is now based in the United States. In addition, India is the world's largest democracy and its growing middle class, projected to grow to nearly 40 percent of its population by 2031 according to Kantar Research, presents an attractive market for multinationals.

GROWTH-FRIENDLY POLICIES

India has been on a firm growth trajectory since the late Prime Minister Manmohan Singh introduced economic liberalization policies in the 1990s. "We are a messy multi-party democracy and even though different political parties have come into power, the overall strategy toward improving economic well-being has remained on course," said Sameer Gandhi, managing director at Omron Automation, India.

That course has accelerated over recent years. According to Deloitte's August 2025 economic outlook, India's growth economy grew 7.4 percent year over year in the final quarter of fiscal year 2024 to 2025. However, understanding that private sector development cannot proceed without infrastructure groundwork, the government has expanded and improved highway networks and port infrastructure. A rebooted Goods and Services Tax (GST) in 2017 allows for a central and less complex tax structure instead of businesses having to negotiate with every state's own chaotic rules.

The much touted "Make in India" initiative, launched in 2014, focuses on increasing manufacturing in the country. The goal is to facilitate growth by offering tax incentives, simplified GST and single-window clearance to reduce regulatory hurdles," said Sachin Wadikhaye, a self-



Skyroot engineers work on integrating the Vikram-1's Orbital Adjustment Module (OAM). Photo: Skyroot Aerospace

labeled innovation evangelist and startup mentor in Pune, India. Make in India attracts foreign investments and has established Special Economic Zones (SEZ) that offer tax exemptions and more liberal business-friendly regulations.

To speed up the movement of research to development, the Anusandhan National Research Foundation (ANRF) promotes research and development and "fosters a culture of research and innovation throughout India's universities, colleges, research Institutions, and R&D laboratories," according to its stated mission.

India's National Highway network grew by 60 percent in the last decade—from 91,287 km in 2014 to 146,195 km in 2024, making it the second largest road network in the world, according to the year-end review of the Ministry of Road Transport and Highways. And the country's National High-Speed Corridors have also increased from just 93 km in 2014 to 2,474 km in 2024. This slate of multi-pronged initiatives is part of the country's push toward Viksit Bharat—a developed India by 2047, which marks the centenary of its independence from British rule.

While these goals are noble and ambitious, they will struggle to see fruition without India addressing one of its biggest challenges: energy security, said Madhukar Sharma, president of ASME India.

The country's import of Russian oil occupied the news in July as one of the reasons for the call for steep U.S. tariffs on India. In what might be a complicated sleight-of-hand, one of India's private sector giants, Reliance Industries, imports Russian crude and sells most of the processed fuel to Europe, Reuters reported. To address the energy gaps, India is focusing on boosting solar energy, evaluating licensing agreements for mobile nuclear reactors, and launching a national green hydrogen mission with production-linked incentives to export green hydrogen to Europe.

Autonomous robotics systems will help India's ability to scale advanced manufacturing capabilities. Photo: Omron Automation India

"Ambitious clean energy targets won't be met without addressing the vulnerabilities in aging power systems and infrastructure," said Unnadha Raja, business head for Sulzer India. Even if the country is making provisions for increased energy security, the energy still needs to be distributed. Retrofitting of shaky power plants and utilities infrastructure is the most efficient way to update the infrastructure, Raja argued. But the tariffs are a prime example of how geopolitics can influence economic growth. While

seesawing tariffs on trade have had multinationals evaluating India as a counterfoil to development in China, Sharma said it's unwise policy to peg growth on such shifting sands. "I'm not a great proponent of a China plus one strategy because geopolitical situations are temporary and India's journey is a little more deep-rooted and permanent," he said.

ENGINEERING AND MANUFACTURING SUCCESSES

Part of that journey is a big push toward increasing manufacturing and the development of home-grown companies in advanced technologies like robotics and space exploration.

Neeraj Sharma rode on the coattails of growth from Addverb, a company that focused on warehouse logistics, to launch Anzo Controls, a Rajasthan-headquartered business that offers comprehensive robotics solutions. He credits the revised GST tax code for making it simpler to do business in the country, especially for the kind of central warehousing that Addverb does. Moving from Addverb to a robotics company was a strategic move for Neeraj as he wanted to be part of a cohort that launched more rigorous automation in India. Having Anzo Controls manufacturing robots in India might have the onshoring effect of easing supply chain pressures as well—troubles that have plagued manufacturing operations worldwide.

India's private sector has ambitions that extend even farther than automation and robotics. Vikram-S, the country's first private suborbital rocket from Skyroot Aerospace in Hyderabad, Telangana, another technology hub, launched in 2022. Pawan Kumar Chandana, Skyroot's co-founder, said the team was confident they could scale quickly to build an orbital rocket after the success of their first mission. In hindsight, it turned out to be very ambitious, he said. Nevertheless, Chandana said establishing a strong in-house manufacturing setup was one of the smartest moves they made. "For Vikram-1, we invested in our own filament winding machines for composite structures, large hot-air ovens and autoclaves for curing them to perfection, and CNC machines for precision parts. Having these cutting-edge capabilities in-house meant we had complete control over the critical structures of the vehicle, could reduce logistical complexity, and cut down delays," Chandana said.

Technicians at Skyroot Aerospace inspect a rocket stage in Hyderabad, part of India's expanding private space sector. Photo: Skyroot Aerospace

On a parallel track, the country is pushing for technology leadership in AI as well and is investing heavily in AI and data centers, Sharma said. In April 2025, the Adani Group announced a $10 billion investment in the country to hit 10 GW in data center capacity. The growth in data centers is going to increase the demand for cooling equipment and that too is creating a demand for steel and a manufacturing push, Sharma added.

BUMPS AHEAD

Before the country achieves the goal of Viskit Bharat, it must overcome a few challenges. India's independent approach to more development notwithstanding, a fair number of experts, including Sharma, hold up China

Ashwini Bhide (center right) leads the Mumbai Metro team on a visit to Alstom India's Andhra Pradesh manufacturing facility, which is producing rolling stock for the Aqua Line project. Photo: Mumbai Metro

as the model to emulate in producing goods in the home country for consumption around the world.China has long been the favorite for contract manufacturing for multinationals, producing goods like the iPhone on spec without owning any of the brand's intellectual property. The country has also made remarkable progress in manufacturing its own goods for world markets.
"For contract manufacturing, China has always been attractive in terms of cost and even quality-wise, it was still better than India," Soni said. And in terms of home-designed and produced goods for the world market, India does need to catch up, he pointed out. Sharma agreed, adding that it's about time we made products in India for the world, but despite investment in research, that has not yet happened.

Anusandhan Research is looking to change that with its advancing research for rapid technology deployment. And the story on the ground is changing rapidly: In late July 2025, India overtook China in the number of smartphones exported to the United States.

While India is gifted with a large population, skilled talent, and low labor costs, it's also marred by low productivity and low-tech automation, especially when compared to China, Sharma said. "Over the past many years, China has invested heavily in research that has led to new materials, new processes, and new equipment. It's truly world-class and the best of the best," he continued.

For India to compete on a global scale, the move to automation needs to happen on a large scale, and it's what Gandhi is pushing through Omron Automation. While there's a uniform awareness that automation needs to ramp up, there is still substantial handwringing about its potential effects, especially in the most populous country in the world with 1.46 billion people.

Youth unemployment in India is high—there are 28 million young, educated people looking for jobs. In addition, while top-tier manufacturers might find the implementation of automation to be smooth sailing, this is not always the case for micro, small, and medium enterprises (MSMEs), Gandhi added. "For these folks, adopting automation might be a challenge because they do not always attract the best skillsets for themselves," he said. Omron Automation is helping these smaller companies level the playing field.

Gandhi is also chair of the Confederation of Indian Industry (CII) National

Committee on Robotics, Drones & Automation 2025-26, an initiative that came to be out of a growing awareness of the need to embrace automation and robotics to become an international player. “There’s a realization that your labor cost arbitrage, which is an advantage in India, is not the only thing that’s going to help. You need to produce things more efficiently. You need to deliver better than expected quality and all of that cannot be accomplished without automation,” Gandhi said.

“As such, we have a lot to do in terms of infrastructure for last-mile connectivity,” Sharma added. While the government has done a commendable job in building highways and private investment has fueled into ports, “the connectivity to the highways from the industrial beds still remains poor,” he continued. Mechanisms such as toll collection, customs collection and clearance, are still rudimentary. “We have to really cut down the time to market or time to ports and be speedy in our shipments,” Sharma said.

With the advent of AI, digital twins, and other tools, there’s also a lot of training that needs to be done, Sharma noted. “It’s imperative for the engineers of the future to be knowledgeable about these tools because their

An engineer works inside the nose cone of a Skyroot launch vehicle, a part of India’s growing space and tech ecosystem. Photo: Skyroot Aerospace

work on the shop floor will not be managing machines because machines will be managed autonomously,” he said.

ENGINEERING A BETTER FUTURE

Raja is excited to see growth on many fronts, including power and utilities and roads and airport infrastructure. It all points to increased momentum for the future.

The startup ecosystem in tech and engineering, now the third largest in the world, is also on a fast track with government initiatives like Startup India, which facilitates easier compliance protocols and tax benefits.

But the big change Bhide has noticed over decades is the civic attitude among citizens. A chalta hai (anything goes) attitude is no longer dominant.

"Indians themselves are eager for change. People want progress and are willing to make short-term sacrifices to get there," Bhide said. And from all indicators, despite the challenges, a combination of widening investments, growing consumers, engineering and tech talent, and favorable government policies, is set to increase at least near-term growth for the country.Many are bullish about India's prospects heading toward the next decade. "In terms of a return on investment, India will be able to counter anyone else in robotics or any other manufacturing. We are getting to a wider scale—it's just a matter of time," Sharma said.

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